ActInf Textbook Group ~ Cohort 1 ~ Meeting 6 (Chapter 3 pt. 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 6 (Chapter 3 pt. 1) | 1:00:31

hello everyone it is june 9 2022

and it's

chapter three

week one

and it's the first cohort of the

textbook group

we're gonna go to the questions

uh there weren't any ideas added for

three but if people have any key themes

or any ideas that they picked up on with

three they can add them here

otherwise we'll go to the

questions and

[Music]

also if people have questions in the

chat or

it would be also great just to take

notes

or add questions and continue to upload

them

and then next week we'll continue the

discussion on three

so

this is just the

opening set of questions

maybe just the first

ideas of the book turning two questions

now that we've

at least

gone through chapter two once and part

of chapter three or the whole thing of

chapter three

what is the high road to active

inference

what does anyone think about the high

road to active inference
or
about this low road high road
construct
that is used in chapters two and three
just broadly how did the approach of
chapter three feel different than the
approach or the topics of chapter two
um jf and then ali
well i i just want to make the bill
comment that the the the the high road
approach is what uh first grabbed me
when i encountered the free energy
principle that that what really
fascinated me that there was an attempt
at the
a unifying answer to uh
self-organization uh
cognition
um
and
that this could all be brought together
uh around some relatively uh
a sparse number of of concepts that's
what grabbed me much more than the low
road
thank you
ali and then ben
well yeah uh reading chapter three
actually clarified uh my previously held
in fact wrong belief that
uh i
previously i thought that a low road was
a kind of
uh bottom-up approach and the high road
is a top-down approach but uh now i see
that
low road was basically

uh tackling the problem of
active inference from the inferential
point of view and from the statistical
and inferential point of view but uh in
the high road
uh we have a much more
general generalized and much more
let's say all-encompassing
uh
view to tackle the problem of active
inference namely
uh
by way of defining markup blankets and
all the properties of living organisms
and so on
awesome thank you
ben and then anyone else who raises
their hand and also you can take notes
on this ideas the book like high road
generally ben and then mike
yeah i think uh
i think for me as well the height this
chapter three in the high road has been
um an easier road in and i the thing
that i noticed about it was it seems to
me so so they say in the introduction
that active inference is a normative
theory i feel like the high road to me
feels there's much more normativity
in the high road is that it's an account
of what organisms have to do and i think
compared to the previous
compared to the previous chapter i think
chapter 2
felt a lot more
descriptive for me i'm not sure other
people would maybe that's just the way
that i was reading it but it

it felt like chapter three was a lot
more kind of it had a more normative
force to it that i i really enjoyed
thank you bye
there we go
sorry
my computer was not responding
uh so i went through a similar
experience as ali described in terms of
going into it thinking about bottoms up
and top down and coming out of it
thinking about it differently
um
but i feel like in chapter three there's
still a lot to unpack around the markov
blanket so
that is still
capturing this inferential nature of
things
it still has a bayesian quality like
what we saw in chapter two
um
but it also for me at least it still
feels a bit opaque at this point in
terms of
um
this sort of indirect influence that's
taking place across the markov blanket
awesome
okay
any other comments otherwise we're going
to go to the questions and and we'll
approach many of those topics
okay
first question
regarding the markov blanket explanation
on page 43 and box 3.1
they write no additional information

about the future is gained by finding
out about the past assuming we know the
present how can this be true that's one
question
they write
they define regarding
the markov blanket as
the set of variables that mediate all
statistical interactions between a
system and its environment
what is a statistical rather than a
non-statistical interaction
they write that they have supplemented
conditional independencies with
dynamical constraints
so that the flows do not depend upon
states on the opposite side of the
blanket
why does independence between the flows
on the two sides of the blanket matter
so much
all right
great so
who would like to give a thought on this
first question how can it be true about
markov blankets that no additional
information about the future is gained
by finding out about the past
so i added a note in there you can tell
me if this is correct or not
with regard to markov chains that as i
understand the markov chain
in its present state is effectively
capturing all the information about the
past so
i
assume that translates into the markov
blanket

yes
um great comment and a markov chain is
is a
uh
used very broadly
and
just to give one example it's like
the current state of the chess board
knowing more about previous moves
doesn't tell you more about the future
and so it's like it's a one markov chain
because the dependency is only about
that one time step now if it were to
violate that condition
of being a one markov blanket through
time
and have like a three markov blanket
that would mean that the same board
position depending on what move happened
three before
was still having some influence on how
the system evolved
that wouldn't violate the markovian
property it would just make it a
different kind of markov chain
and that is commonly used in the
temporal modeling like in time series
modeling
and in
like transitions of stochastic processes
the markov blanket is abstracting it
from
this like time sequence idea like the
now is a blanket between the past and
the future
and we could think about spatial systems
um
like a newton's cradle or something like

that
as well as like spatial temporal systems
or just abstract causal systems
um so how can this be true it's true by
definition because that's what the
definition of a markov blanket is
whether that description
is adequate
or maps on to carving nature of the
joints and all these other things are
um respectively
statistical model comparison and
ontological philosophical questions but
it's true by definition about markov
blankets lyle
yeah i mean i think you actually you
just pretty much covered it but i would
just say this is the same condition if
you're building rl models of you know
different kinds of autonomous systems
this is the same restriction
i think your comment about the nested
the nested
blankets is interesting because
in the rl space you know we we typically
don't have that kind of richness but
what we do then would be
for example we want to know that just
and in that case we would just create a
you basically create a state variable
which is the
the pro you know the
the history of the
uh of the state transition so we fudge
it a little bit but your your
explanation that
in in
in this conception with nested markov

blankets then that that would be the
more comprehensive solution to that
issue
thanks just to kind of clarify there
like one hack or work around would be
like you make the list of all of the
last possible three moves
and then you make a one markov blanket
with all the last combinations of three
so it still uses the machinery of a one
markov chain which has the simplest
transition matrix
but it could encompass the last states
but it sort of compresses them into a
one markov chain format
yeah i mean very often you want to know
the you know cross you want to know the
lat the average of the last you know 20
time steps yeah you're just trying to
smooth something right
and it's just way too much work to build
all that machinery in there so you
you make you make that average actually
a present variable is what you do
great example if something really does
depend on the last 20 time steps or you
want to do model comparison as to
whether it's a better model that is
conditioned on the last 15 or 20 or 30
um you can just
condense it with summary statistics and
descriptions again into this one markov
framework otherwise the combinatorics of
how the 20 influence each other is vast
brock
i guess that's related to it like my
comment was just going to be that um the
way they worded it

if you're not
rigidly thinking about us
singular markov blanket
that
you know i mean
if just just practically speaking
finding out about the past
usually does give you information about
the future if you're talking about
outside of a singular marketplace right
so um
if you you know adding the words um
no additional information like uh about
the mark of you know about the future of
the markov blanket is gained by finding
out about the past the markov blanket
like kind of you know bounds that but um
in the case of like the machine learning
example and comparing models you're
talking necessarily talking about a much
larger archive blanket than just a
singular one there right so
that's just the same thing
yes and just like many concepts have
like a broader
conversational informal sense and a
narrow more technical sense assuming we
know the present means fully knowing the
presence fully knowing the blanket so
it's like but i could still learn things
about the past so that's not to say that
there isn't novel information that you
could discover about the past like you
could still go back in that chess game
and learn information so it's not that
there isn't information in the past
it's not that it wouldn't even be
interesting it's that for the purposes

of how the present goes into the next
time step
the present blanket in this temporal
markov chain which is like a blanket
through time
it contains the information you need
for the transition frequencies
um okay
they define that it's the set of
variables that mediate
all and also the markov blanket um
using last names
you know
personal opinion is not
helpful because not only are there
multiple markovs but the markovian
property is not something that's
technical or defined
so
maybe
someday we'll have better ways to
describe and be more specific that don't
involve invoking
ambiguity around names
that being said
it was worked out analytically by
markov and sun
and others
but most recently it was pearl 1988 and
causal inference with bayesian
perspective so this is not like an act
of inference ism it's drawing upon a
total
bayesian statistical framework
any base graph that's not fully
connected
is going to have some nodes
that intermediate so that's going to

relate to this question
so what is a statistical rather than a
non-statistical interaction
ali and then anyone else who would like
to address that
well i think it has something to do with
the way uh these interactions are
parameterized and modeled because you
see every phenomenon
can be
parameterized and modeled uh from
many different many various perspectives
and
frameworks within various frameworks
but here and the reason i think they put
the term statistical in parentheses is
that they mean that
these variables mediate all interactions
that have been parameterized
statistically
so the i think it's
it refers to the nature
of
parameterizing and modeling these
interactions
as opposed to
distinguishing between inter statistical
and non-statistical
interactions
all right
so anyone else can raise their hand but
just here's a few thoughts on that
um a statistical interaction is the edge
in a causal bayes graph
and people might be more familiar or
have seen like structural equation
modeling where nodes are variables and
edges are the correlation

between or among variables
conditioning on the structure of the
graph
um in the causal bayes graph the nodes
are variables and the edges are
statistical causal influences so
causality and statistics is not always
the same as what people mean by cause in
the real world
but like granger causality
and just this notion of like coarse
graining and cause
it may line up
with
the sort of narrative natural language
description that somebody has
between
when they're speaking it might be
narrower than that it might be a
different thing than that it might be
more general than that
um would a non-statistical interaction
be i mean
there's various ways to approach that
it's kind of pointing towards some sort
of
like a touch could be an interaction
but what if things are touching but
those variables don't
change as a function of their touching
then do those things interact or not
that's one question
and then cause there's like just
tremendous
real world and philosophy questions like
what causes what what's the difference
making cause what's the cause that makes
a difference

necessity insufficiency
there's so many
questions in this like causal philosophy
and this is highlighting
we're talking about the model
we made a base graph with height weight
and shirt color
and here are the statistical effectors
and so we're not saying that there's no
interaction between this and that in the
real world now we're within the model
and we're talking about the causal graph
and again unless the base graph is fully
connected
there's going to be some
partitioning scheme that will result in
some set of nodes
being conditionally independent from
another set of nodes
when conditioned upon a blanket so it's
not like features of the real world or
even variables in a statistical equation
can just be unilaterally tagged as
blanket states
it's always a partition that
co-instantiates
two sides
internal and external but there's a
symmetry
and the blanket
so it's not like features of the world
or even features of the model are
intrinsically internal external or
blanket states
but that's a partitioning scheme that
can be applied
to base graphs of like a vast
various structures of bayes graphs

and bayes graphs can be applied to a
vast number of real world situations
and many many papers and live streams
discuss like sort of like
all-encompassing markovian modeism
markov blankets all the way down nested
markov blankets are the world
ranging to the critical perspectives
um
not necessarily detractors but just
those who believe that the usage of the
concept is uh
out of alignment with ways that people
have used it
mike and then anyone else
so does it make sense to think of a
markov blanket
in the context of an internal agent
state and an external environment is the
markov blanket serving as this kind of a
decoupling component because of
the way that things are conditionally
independent
based on the variables inside the
blanket
what do people think
yes
i mean for the internal state to be
different than the external state or
have some other equilibrium other
non-equilibrium state that the external
environment is is in
you would have to necessarily have
i'm not sure if decoupling is the right
word but some sort of conditional
dependent independent sort of
zone that separates those two regions
so here's figure 31 thank you brock

the active and the sensory states
compose the blanket
now
in various like live streams and papers
the history of the concept again from
the analytical pre-computational
phrasing
of insulation of fluctuation of random
variables
to pearls bayesian and computational
framing however pearl's markov blanket
concept doesn't have a delineation of
active and sensory states
it's only like there's one kind of
fabric in that blanket it's just blanket
states
fristen
maybe this is also will be shown to be
traced to other places and ways but
one of the core things that fristen at
all have brought into the picture
was interpreting within the blanket
states
the states that have incoming
dependencies to the system of interest
as sensory
and then the states that have outgoing
dependencies from the system of interest
as active states
so
people may already see
like challenges and complexities that
arise
with the arrows and the directions
and a lot of that is addressed in the
how particular is the free energy
principle
of aguilera at all

and um
so
we're going to continue talking about
this
entity partitioning
ali and then lyle
uh well i think uh we should point
to a very
important typo typographical error in
this
picture here because uh in this figure
active states are expressed in terms of
external states and markup blankets and
sensory states are expressed in terms of
internal states and markup blankets
but it should be vice versa i mean
the flows of internal active states are
independent of external states and the
flows of external and sensory states do
not depend on internal states so we
should change
x and mu in those
two equations
yes sometimes with a um possible
typo like that it's like it's it's so
egregious slash blatant
that it's challenging to even know but
we can absolutely um
ask the authors but just to like read
the equation and think about why it is
that way
the dot is the rate of change of a
variable so the internal states are mu
and so this is the rate of change of
internal states rate of change of
external states x and the rate of change
of u the active states and y the sensory
states so we're looking at how these

partitioned states in a bayes graph
change through time
the states that are the blanket and the
internal
are the particular states
those are referring to the particle
that's like figure versus ground the
particle that's moving around the
curious particle the active entity
is the particular states that is
partitioning the thing
literally the thing
away from the niche
then there's the blanket states that are
intermediating that interface and then
we'll also talk about autonomous states
which are just the internal and the
active states
so the equation is saying the rate of
change in for example active states
is a flow function f and a noise
function
both the flow and the noise function the
subscript is referring to like that one
it's kind of like ipso in latin like
it's the flow on mu and the noise on mu
the flow
which is what the principle of least
action converges us towards
in the limit when the statistical
fluctuations from the omega are low
is a function of
certain variables
and so this is saying the flow
of active states
is a function of
external
sensory

and active states
however as ali has pointed to
the arguments that should be
guiding the flow of active states
are actually the blanket states and
internal states
not external states and then analogously
sensory states should have a flow that's
defined in terms of blanket states
and external states
not internal states
um lyell and then ali
yeah so actually i think you're getting
right right at the question at hand
because i was confused about the the
bi-directional arrows particularly
between internal states and active
states right
and
and i think
you're drilling right into that i still
don't
completely understand what that picture
should look like
uh but i i have trouble
conceptually i sort of get a conception
i feel like i get the markov blanket but
that representation i have trouble
working through the details on that
cool so um guest stream number seven
how particular is the physics of the
free energy principle
it explores different
um
[Music]
entity niche causal relationships
like for example we could imagine
a simple around-the-clock

entity model
external states induce sensory states
one directional arrow
sensory goes into internal internal to
active active to external like an ooda
loop or something like that but just in
around the clock no bi-directionality no
connection between active and sensory
but
given this partitioning one could
imagine that topology of cause
one could imagine all kinds of
topologies of cause some that violate
the markov blanket condition
like internal states and external states
that are um
no longer internal and external because
they would have a causal link but even
preserving the markov blanket condition
we can imagine different
topologies different sparsities of
couplings between these different states
a bi-directional arrow is kind of like
there would be a non-zero cause
and in the other side of the matrix
it would also be a non-zero cause
versus a unidirectional relationship
where like variable two has a causal
influence on three but three doesn't
have it on two
in this paper um in guest room 7.1 and
and that paper and further line of
research
they explore
what
sparse coupling structures
causal architectures of the entity niche
relationship and interface

would grant different statistical
capacities so
um i could be wrong on this
but active inference is not a commitment
to this specific
coupling architecture
it may be possible to design or describe
or imagine systems
that have different causal architectures
maybe even
true at this basal most kernel loop
but certainly true when we start
thinking about nested structures
and cognitive structures and so on
um so yes it is a little unfortunate
about this
however um
like
this is just thinking about causal
relationships
of the particular partition
and we're describing statistical
relationships here causal influence as
inferred from like time series data with
grainger causality or other methods
on specific random variables
so this is not the conversational notion
of cause like but what i'm thinking
is eventually resulting in things
happening in the outside world
or aren't things in the outside world
eventually changing things yes
but just like the chessboard through
time
the blanket
is statistically
intermediating and again that's why it
gets so messy

with the application of like well the
retina must be the sense
states
if one can read that example
and
not fall into the quicksand it could be
didactic
but
to tag retina as sense state
is like the tip of the iceberg of
a partially specified model
often one where the measurements have
not actually occurred and so on so then
that can be extremely difficult for
those with less familiarity with the
formalism to generalize accurately from
because shouldn't it be like you know
retina sense states and then like the
arm is the active state but
it's a partitioning that co-instantiates
and it's model dependence it's not
describing features of the real world
ollie
uh yeah i just wanted to point uh
one of
uh
ram said at all's um recent
fascinating paper on bayesian mechanics
which i put the link
in the chat and in that paper especially
in section three
they have a very
illuminating discussion about these
whole business of uh partitioning and
markup blankets and uh one sentence
that's uh
pretty relevant to our discussion here
is that the key point to note is that

the flow of internal and active
components
i.e their trajectory through state space
does not depend upon external components
and reciprocally
the flow of external and sensory states
or paths
does not depend upon internal states or
paths from page 15. so
yeah in that
if
you refer to that article
it has been elaborated much explicitly
also
in four days we will have
dalton and maxwell at all for a guest
stream
the paper just came out now we'll be
able to
have a discussion with them and um
learn and ask them questions so like if
people have suggestions for guest
streams
that can be accomplished if people want
to facilitate or participate in
different streams if they have questions
for different streams
being there live but especially getting
involved eagerly and actively and early
is the biggest leverage point
because that allows us to design the
material so that it's like permanently
useful
rather than potentially
ad hoc questions that arise during which
can be super important and um hopefully
people can see
some affordances for participation and

development
um let's just get to this last question
and it's it's like we have other
questions but these are essential and
it's one of the
opening formalisms of the chapter
um
just
to conclude here though why does
independence between the flows on the
two sides of the blanket matter so much
these are good things to
think about um just one short thought
would be without independence between
the flows
there is no distinguishing one thing
from another
ali
and also i think it relates to the uh
what we uh read in chapter two about
hidden states because uh that's what the
word hidden probably refers to i mean
internal state does not have uh direct
access to external states and vice versa
so
that markup blanket pretty much
formalizes uh this hiddenness
yes
awesome and um partially observable
models of the bayesian type
which are used in like partially
observable markov decision processes
expectation maximization any kind of
bayesian priors and hyper priors and so
on
all of these
have bayes graphs
that reflect conditionally independent

variables
so it's not that they're independent
in a conversational way like they don't
have any way of communicating
it's that they are specifically involved
in a causal network that has conditional
independence
but we'll hopefully develop more answers
and notes here we'll move on
what is the um differences between and
among the terms bayesian brain
hypothesis
predictive processing predictive coding
and active inference might you be able
to suggest references or resources for
where these distinctions are delineated
sure
so anyone can add more but here's i
think two key resources here
the first is livestream 43.
in 43.0 maria gave a really excellent
overview
of
predictive processing and predictive
coding from a historical perspective
and briefly she proposed that predictive
coding can refer to a uni-directional
data encoding scheme
of the kind that we see in signal
processing video compression so on
whereas predictive processing is
referring to a bi-directional
architecture where predictive coding is
implemented
in this ongoing top-down bottom-up way
bayesian brain hypothesis was also
touched on and we connected this to
active inference but 43.0 is a long

technical review paper but we have three
discussions on it predictive coding
theoretical experimental review
recent paper
very um good source for the um
predictive processing and coding
and predictive processing and coding
initially was more of about a sensory
framework however
at the end of that paper they show okay
now we're going to do predictive
processing on action it's active
inference
so predictive processing about action
is
six of one and half dozen of another
within a margin of reasonable
approximation
similar but there's also some key
differences
like
reliance on different
formalisms but these are all things for
us to explore and unpack as we work on
the ontology and so on
um
specifically
predictive coding active inference in
the bayesian brain
this is an interview that
a departed colleague
and i
did with carl fursten in 2018
and we specifically asked the bayesian
brain hypothesis predictive coding and
the free energy principle are often
equated with one another
you yourself has suggested that the

three frameworks are variations of the
same basic mechanisms
so 2018 was a very different time
active inference had not been delineated
from
fep in the same way that it is coming to
be now and there was a lot of other
differences
between that and now but he gives an
extended answer
so for people who want to learn about
that his extended answer
and martin's restatements of the
bayesian brain hypothesis
are still some of the best places to
find clarity on that issue
does anyone have any other
thoughts or questions on this
just one other note while anyone can
raise their hand is like the bayesian
brain hypothesis
sometimes um
[Music]
could be used in a a very
um instrumentalist way
we're using bayesian statistics to do
neurobehavioral research
it might be
um a realist implementational
um claim like the brain is doing
bayesian statistics or something like
bayesian statistics so this hypothesis
floats kind of among the layers of mars
analysis
like implementational algorithmic
computational
um
[Music]

and people do use it to mean different
things
and in bayesian graphs in bayesian
statistics it can be about perception or
it could be about action so bayesian
brain hypothesis depending on how
somebody frames it
and what specific models they're talking
about
might include
for example all of these because these
might be
applied in a bayesian framework to talk
about the brain
ben
yeah i mean i think you've covered it
really really well but i i wish i'd have
had access or knowledge of the result
the the
talk that you just mentioned because
trying to differentiate predictive
processing from active inference has
been a kind of source of pain and
confusion for me over the last few
months and
i just i thought another way of putting
it perhaps because i think i know the
difference now is essentially
um
some
presentations of predictive processing
are active inference
in the sense that they're highly
embodied inactive
embedded
and so if you have a predictive
processing that draws on these four
reaccounts of cognition it tends to just

that just is active inference active
inference is kind of necessarily
inactive and embodied and there's a
paper um
i should say there's a paper by i i
believe it's called wilding the
predictive brain
it's by andy clark kate nave
mark miller and
george dean i think which basically
takes predictive processing as a theory
of
neural processing and weaves in these
kind of
cognitive these four e narratives really
really well and kind of fleshes out a
much more
active infancy picture it's kind of the
paper brings predictive process and
active inference together really nicely
i think
thanks great suggestion and
just without going into too much detail
if this is the causal graph architecture
of a predictive processing system
one can imagine that like mu
is
being a markov blanket with respect to
these two epsilons or so on
so predictive processing doesn't
highlight as much the markov blanket
formalism predictive processing is not a
physics for particular systems
like fep is
but
it's not incompatible
so
maybe they're just different fingers

pointing at similar or different parts
of the moon
um so there isn't an incompatibility
especially as over the recent years
active inference implemented predictive
architectures and approaches
and predictive processing started to
undergo that pragmatic inactive 4e5e
whatever
um infusion
which led to them modeling action as a
variable
and so all they did was they just made
it so instead of the free energy being
only about sensory observations and
expectations they just tucked action in
and it doesn't radically restructure the
architecture
it just means that action is a variable
in these equations
so
um
43.1 uh zero one and two are all good
places to look
okay
um
this was a short question internals
estate and external state were added to
the ontology but sometimes um like if
you have a quotation mark and then an at
it won't bring it up
so sometimes you have to type it with an
at first
okay
in our um
last
15 minutes
okay these two are um good maybe we

could do a first pass on them and then
in the coming week like add notes and
more questions to discuss
they wrote if one defines preferred
states as expected states then one can
say that living organisms must minimize
the surprise of their sensory
observations
i'm not clear on the ontology here
is it that preference is the same as
expectation
or that formally we can use them in the
same equation or something else
does active inference make ontological
claims like
preference is actually expectation just
as heat is actually the excitation of
molecules
more generally is the free energy in
physics just an analogy or is it an
ontological assertion
who would like to approach one of the
aspects of this question
i can i i would like to take a stop but
again i guess guess yes please
i guess
it's not saying that they're the same
but that
we want like if we're minimizing free
energy
we're trying to make it the same
and but if they're surprised they won't
be the same
i guess the goal is to get them as close
as possible
that's that will be my guess on this
thank you
good insight um also point to two ways

that expectation is conversationally
used
um the expectation with the fancy e
is the expected mean of a distribution
like expected returns might be about the
future but it's actually referring to
the mean estimate of a distribution in
the gaussian case the mean and the mode
are the same
mode seeking and mean seeking are
equivalent
there can also be a variance estimator
and so on so the expectation of a
gaussian is like tremendously
informative
expectation can also mean
things that are
predicted about the future
that's a conversational what do you
expect it to be tomorrow
both can be conjoined like
what do you expect the temperature to be
tomorrow
is the expectation of a distribution of
the uncertainty of the temperature
distribution at the time point tomorrow
so they're not exclusive definitions
it just has to be understood
what this means
because it'd be like well if our
predictions about the future
are our preferences you know then
there's all kinds of tangles that you
know one might find themselves in
also
um
the um
i'm uh not remembering which number it

was
but um
[Music]
the dual usage
of the same term
to reflect the organism's preferred
states
and also
the centering of the distribution
[Music]
at that time point and other time points
and then just like jessica mentioned
this
operation of free energy minimization
is in service of reducing the divergence
between the preferred state
and the um
the expectation of the preferred state
which is can be set or learned
and then the um
the observations that are coming in
so
um
in one equation it leans more towards
the english description of a preference
like the organism prefers to be at 72
and in the other sense that is like the
expectation of a distribution
and there's some observation
distribution
with a minimization of a divergence
such that
if
the observations
were realizing preferences
then those distributions would be
aligned
um

so yes
these terms are not
these terms are more like phenomena or
natural language tags
also it'll be awesome in
the last
two weeks of june
in 46 active inference models do not
contradict folk psychology
this will be extended extended
discussions on like wanting beliefs
intentions
which variables inactive
are appropriate
for being described that way
does it contradict one or the other
those are like things that we'll explore
so
the equations are just what they are
and then
in some usages
a variable is like
the organism's preferred states what
it's trying to reduce its divergence
between
and also that is like where it expects
to be and that's what licenses the use
of surprise
and minimizing surprise
because
an observation right at the center of
the gaussian is minimally surprising we
talked about that in chapter two an
observation that's outside of the
gaussian is highly surprising
so
if we have some sort of distribution
of

outcomes
we could ask how closely are they
aligned with our preferences
in terms of surprise
due to this dual use of the same
variables
does active inference make ontological
claims do theories make claims
do people make claims
eg preference is actually expectation
just as heat is actually excitation of
molecules
anyone can raise their hand at any time
we'll
talk about that in the folk psychology
but it's an interesting question ben
yeah
i wonder if um it seems to me like these
questions are taking us maybe into the
map territory um debate that we i think
we spoke about last time
about the you know how um
realist or concrete should we think
about
um
the kind of ontological claims of active
inference there's certainly certainly
the last part of that question i think
about whether we should take it as an
analogy
um
so yeah i think it's quite a big
question but i don't i don't know
um
okay i won't find the slide here but um
is the free energy in physics an analogy
not sure if this is saying like the
gibbs free energy or if that's actually

talking about like the variational free
energy or the expected free energy which
are statistical so
i don't know if it's an analogy
or an ontological assertion
other than
um
just to connect it to like a linear
model with a gaussian error distribution
is that an ontological assertion that
there is a gaussian error distribution
in the world
no
so
are statistical quantities ontological
assertions about the world
maybe there's some nuance but broadly
speaking no
um
and also just like on the math learning
group we've
expanded our operation
due to many
active participants
so just note that the
meetings are at 19 utc on monday tuesday
wednesday and thursday
and they're in the discord voice chat
not in this gather
but
um
we've been working on the resources
on the notation
basic and background questions as well
as overviews of the math for different
chapters like basically just summarizing
and then
um

several um people and this is an example
of like the broken link it's like
variational free energy is deleted so
just one time somebody has to come
through
and just add it back in
but these are natural language
descriptions
that will be very easy to
um transmute into computer code and
translate amongst human languages and
provides a lot of legibility and
comprehensibility so
a lot of the math group has been
modifying like the equations
and everybody is welcome to join those
like discord sessions
and contribute with questions or with
knowledge and expertise like no matter
where somebody is in learning or
discussing these equations
they are the essence of active inference
so questions random notes related
thoughts expertise are all essential for
us improving our shared understanding
and then just in the last five minutes
um page 42 they wrote
in advanced organisms preferred states
can also extend to learned cognitive
goals and go on to say that advanced
organisms like human
can achieve
preferred states
by increasingly abstract social cultural
strategies like they talked about
thermoregulation
and then all the way up to like air
conditioning distribution systems

my question is
if they are cast in terms of active
inference
must all of these preferred states and
strategies be ultimately linked to
survival in some way
or in the case of advanced organisms can
free energy be related to something
other than survival
mike and then anyone else
yeah it feels like this notion of
survival and preferred states gets
conflated in a
sort of odd way
not everything is about survival right
and so
you might have
some measurement of surprise if you're
in a temperature that's 120 degrees but
it's survivable for a period of time
whereas if you're in a temperature of
300 degrees fahrenheit
not survivable right
um
but if you contrast that with other
things that could be equally surprising
or outside the distribution as you
described it earlier
the notion of that all swans are white
but then you find a black spawn and so
uh you'd never conceived that there
might be a black swan before that lives
outside
your distribution and creates a surprise
but it's not necessarily survival rated
thank you so there's a there's a sort of
a missing severity in there somewhere
okay thank you ali and then lyle

well i think work life here is used
somehow in a more black way
meaning system that resists dispersion
and energy dissipation
so it's not necessarily biological
survival
yes great point systems that fail to
persist survive will not
continue to be that thing
and that's just in like a physical sense
repeated measurements are only enabled
by systems that are persistent at that
time scale not eternally
and so when we want to have a theory or
a framework or a physics for things
then
they have to look as if they are
minimizing free energy otherwise they
will simply not be that thing
ben and then lyle
yeah so this was my question i suppose
just to just to kind of clarify it
quickly um
it just one of the things i wonder most
about active influences it seems to me
at least in my life i'm quite often
making decisions and having preferences
that seem so abstracted from
you know my attunement with my
environment or my continued viability as
a system you know like my preference
which mug i'm gonna drink coffee out of
in the morning
it seems
in some sense removed
from
my free energy the way that free energy
is described in the literature um

so that's kind of what i meant
great in um
uh yesterday's 45.2
we asked specifically like how can all
behavior be optimal when like mistakes
are made or oddly ask the question like
what about when there's like sacrificial
or alt altruistic behavior
and there's
a great answer in 45.2
but suffice to say that with different
priors all behavior can be cast as
optimal behavior so the normativity
isn't referring to what should be done
or what is best but it's rather like a
process normativity lyle
right so yeah
this for me tees up the question i
continue to have about the approach
which is how does how does the
concept of novelty
which
some
organisms would
pursue
fit into
this concept of i mean how would it even
be represented here and and novelty is
not just a human trait but you would see
it across all kinds of creatures
that would seek out would seek out novel
and surprising experience you know not
expected
so i for me it's it's kind of an open
question about how that's captured in
this framework
yes the beginning and ending of 45.2
carl's um focus was on curiosity

and
there's various ways to approach it and
um
[Music]
it's an actual question about novelty of
different types and scales
and
within the hierarchical modeling one can
understand novelty of different kinds
and this again is just like the kernel
this is like the single linear
regression
and then there's like model selection
across hierarchically nested linear
models
much development beyond the kernel but
this is like a kernel but um and
especially when thinking about expected
free energy
a path can move to a novel area
because it's part of an expected
trajectory that does reduce risk and
ambiguity and so on so
that's the deflationary approach to
novelty pursuit
jessica and then we'll end the
discussion
at the end
danny you mention it and i guess i i see
novelty
i sort of like the epistemic origin
and so it's like you are
seeking on like new information new
things
in order to minimize uh that uncertainty
i not so much that novelty is on cellar
than t because like like if we look at
um

no i've been thinking like oh it's new
and therefore it's going to be
surprising
and and so we want to minimize those
things and then yes or also if we look
at uncertainty equal to risk we can say
like novelty
is because of the unknown represented
risk and if we're minimizing uncertainty
we're trying to minimize rest
but from what i saw a little bit on
chapter two
and it's sort of like we're not
necessarily trying to minimize race or
on
like the unknown
and like depending on our preferences we
might tolerate high res
and what we're trying to minimize is the
surprise of what we expect like if i'm
in finance
and i expect very high risk in an
investment but i already know that
and
you know and high risk comes back on
like my um uncertainty was minimized
even though there was a lot of risk or
there were a lot of like unknowns on or
like novelty and things like that but it
was already in my expectation of what
was going to happen
so
i was already minimizing that even in
reality it may look like um
that there was a lot of uncertainty uh
in actuality
um but in my model
there isn't and so that's kind of like

how i explained that and to me a little
bit and in terms of novelty for like
adventure and like
on curiosity and things like that i i
equated more to epistemic foraging as a
way of learning and updating the model
and that's why we're doing it
thanks jessica i'll just add one closing
note on that
i would argue that it's the
pragmatic
absolutism the reinforcement and reward
learning centrism that has curtailed an
effective theory of novelty or curiosity
because
radically different kinds of actions
have to be coerced into a value
framework whether it's deep q-learning
or reinforcement or anything like that
there has to be like a novelty bonus or
all these other ad-hoc unprincipled
approaches they might be super technical
they might be super effective but
they're not driven from first principles
in variational free energy f and
expected free energy g
there's this like minimum of two
with pragmatic value which is to um
bring alignment between the preferences
and the observations
and the epistemic value associated with
what is casually called novelty or
curiosity at that scale of potentially a
hierarchical model so active inference
does provide partitioning and an
imperative
that helps us address the phenomena of
curiosity and of novelty in a way that

is quite disparate from trying to coerce
it into a value of curiosity is
expected pragmatic value or the value of
basic research is the expected utility
that it could bring
so but these are all really awesome and
open and ongoing questions so
we will um close the recording and then
have just a one minute break then in
this room we'll continue with dot tools
organizational unit and if anybody wants
to like talk about something else they
can go to a different um gather room so
thanks everybody and see you next week
thanks